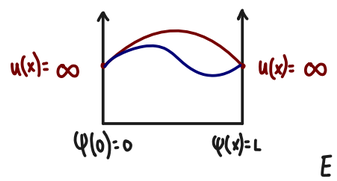
## particle in a box



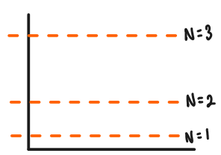

from last frame:

$\lambda = \frac{2L}{N}$ eigenstate

Schrödinger:

$$E = \frac{p^2}{2m} = \frac{(\hbar k)^2}{2m} = \frac{\hbar}{2m}\left(\frac{2\pi}{\lambda}\right)^2 = \frac{\hbar^2\pi^2}{2mL^2}N^2$$

## natural units:




mass of electron: $mc^2 = 0.5\times10^6$ eV ← unit of energy

$\hbar c = 2\times10^{-5}$ eV-cm , $1\text{Å} = 10^{-8}$ cm

$$E = \frac{(\hbar c)^2}{2(mc^2)}\frac{\pi^2}{1\text{Å}^2} \Rightarrow \frac{(2\times10^{-5}\text{ eV-cm})^2}{2(0.5\times10^6\text{ eV})}\frac{36}{10^{-8}\text{ cm}} = \frac{4\cdot36}{1}10^{-10}10^{-6}10^{16} \approx 36\text{ eV}$$

harmonic potential: $U = \frac{1}{2}K^*x^2$ ← spring constant

$F = -Kx$

Kinetic energy: $E = \frac{\pi^2\hbar^2N^2}{2mL^2}$ , approximate: $E \approx \frac{1}{2}KL^2 \Rightarrow L = \sqrt{2E/K}$ ← effective length

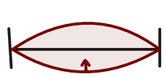
$$E^2 = \frac{\pi^2\hbar^2N^2}{2m(2/K)}$$

solve for E

$E = \frac{\pi}{2}N\hbar\sqrt{\frac{K}{m}} = N\hbar\omega_0$ = exact : Particle energy

$\sqrt{\frac{K}{m}} = \omega_0$

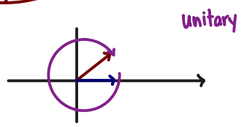
photons, phonons } "bosons" obey standard wave equation $\frac{d^2\psi}{dt^2} = v^2\frac{d^2\psi}{dx^2}$



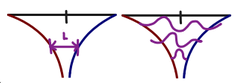
$\psi(x,t) = 0$

unitary evolution

electrons, protons, neutrons } "fermions" obey Schrödinger equation $i\frac{d\psi}{dt} = (\quad)\frac{d^2\psi}{dx^2} + \text{const.}$



$\frac{1}{r}$ potential



$e^{i(\phi)}$ ← phase

$\Delta\phi = 0$

$\Delta\phi = 2\pi = 2\pi R = \lambda$

boundary condition for circular path: $\lambda = \frac{2\pi R}{N}$

$V(r) = \frac{-e^2}{4\pi\epsilon_0 r}$ , write in terms of $\frac{1}{R}$: $\frac{1}{N^2} = \left(\frac{\hbar^2}{2m}\frac{4\pi\epsilon_0}{e^2}\right)\frac{1}{R}$ = $v(r)$:

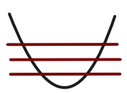
equate potential energy reduction with the kinetic energy gain

$$\frac{e^2}{4\pi R} = \frac{\hbar^2}{2m}\left(\frac{2\pi}{\lambda}\right)^2 = \frac{\hbar^2}{2m}\left(\frac{2\pi}{2\pi R/N}\right)^2 = \left(\frac{\hbar^2}{2mR^2}\right)N^2$$

($\frac{2\pi}{\lambda} \to k$)

"second" quantization
- where do particles come from?

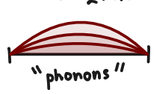
guitar string



$E = N\hbar\omega_0$

$V = \frac{1}{2}Kx^2$

"first quantization"
wave length $\lambda$ is quantized

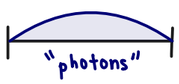


"phonons"

quantized amplitudes

"quantum field theory"

"optical cavity"



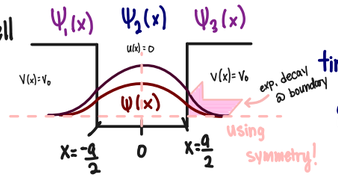
"photons"

infinite # of wavelength modes: $K, \lambda$

↳ for each $\lambda$ - infinite # of possible amplitudes

$N = 0, 1, \ldots$

ex.) finite square well




time-independent Schrödinger

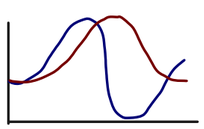
$$E\psi(x) = -\frac{\hbar}{2m}\frac{d^2}{dx}\psi(x) + U(x)\psi(x)$$

$E = i\hbar\frac{d}{dt}$ , $E = \hbar\omega$

$\psi = \psi(x)e^{-i\omega t}$

guess $\psi(x) \propto e^{iKx}$

↳ $E\psi = \frac{-\hbar^2K^2\psi}{2m} + U(x)\psi$



↳ $\psi_2(x) = Ae^{iKx} + Be^{-iKx}$

$\psi_1(x) = a_1e^{Kx} + b_1e^{-Kx}$

$\psi_3(x) = a_3e^{Kx} + b_3e^{-Kx}$

$\psi_1(-\frac{a}{2}) = \psi_2(-\frac{a}{2})$

: $\psi_2(\frac{a}{2}) = \psi_3(\frac{a}{2})$

$\left.\frac{d\psi_1}{dx}\right|_{x=-\frac{a}{2}} = \left.\frac{d\psi_2}{dx}\right|_{x=-\frac{a}{2}}$

$\left.\frac{d\psi_3}{dx}\right|_{x=\frac{a}{2}} = \left.\frac{d\psi_2}{dx}\right|_{x=\frac{a}{2}}$ ⇒

energy equation:

$E = \frac{\hbar^2K^2}{2m}$ in middle region

$\psi(x) \propto e^{iKx} = e^{i(iK)x} = e^{-kx}$

$$K = \sqrt{\left(\frac{2mU}{\hbar^2}\right) - \frac{2m}{\hbar^2}\left(\frac{\hbar^2K^2}{2m}\right)} = \boxed{\sqrt{\beta^2 - K^2}}$$

($\beta$)

↳ special property:

(from symmetry) → $B = \pm A$, $b_3 = \pm a_1$

unknowns: $\psi$ amplitudes, $K, K$

leave overall amplitude free $A \rightsquigarrow A = 1$

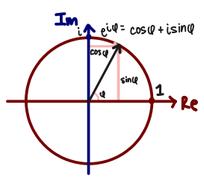
(additional constraint: "normalization")

⇒ boundary @ $x = \frac{a}{2}$: $\underbrace{\left[e^{iK\frac{a}{2}} + e^{-iK\frac{a}{2}}\right]}_{2\cos(\frac{Ka}{2})} = b_3e^{-K\frac{a}{2}}$

$$\left[(iK)e^{iK\frac{a}{2}} - (iK)e^{-iK\frac{a}{2}}\right] = Kb_3e^{-K\frac{a}{2}}$$

↳ solve for symmetric case first: $B = A$

$e^{i\phi} = \cos\phi + i\sin\phi$




$* \; iK\underbrace{\left(e^{iK\frac{a}{2}} - e^{-iK\frac{a}{2}}\right)}_{2i\sin(\frac{Ka}{2}) = -2\sin(\frac{Ka}{2})}$

$$\Rightarrow \frac{+2K\sin\frac{Ka}{2}}{2\cos\frac{Ka}{2}} = +K$$

($\sin\frac{Ka}{2} = \tan\frac{Ka}{2}$)

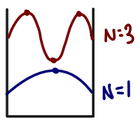
$K = \sqrt{\beta^2 - K^2}$

high K = high K
↳ large kinetic energy
higher vibrational frequency

* solve for k

→ $E = \frac{\hbar^2K^2}{2m}$

↳ $K\tan\frac{Ka}{2} = \sqrt{\beta^2 - K^2}$




↳ $U_0 \to \infty$, $\beta^2 \to \infty$ } $K\tan\frac{Ka}{2} = \infty$, $\tan\phi \Rightarrow \infty$

when $\cos\phi = 0$ @ $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$ :

→ $\frac{Ka}{2} = \frac{\pi}{2}N$ ⇒ $\frac{2\pi a}{\lambda} = \pi N$ ← odd

$= \lambda = \frac{2a}{N}$